

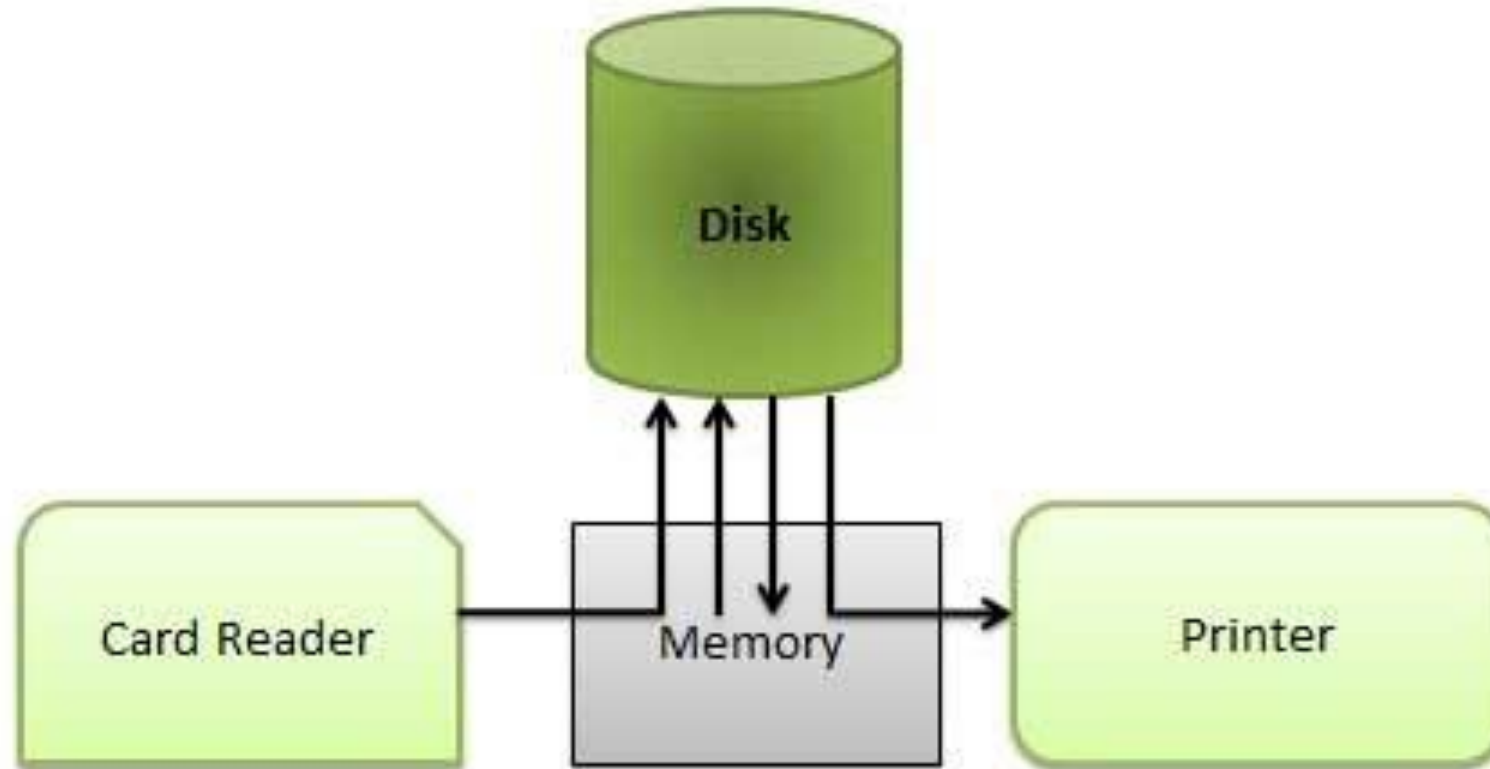
Spooling and Buffer

- Spooling is an acronym for simultaneous peripheral operations on line.
- Spooling refers to putting data of various I/O jobs in a buffer.
- This buffer is a special area in memory or hard disk which is accessible to I/O devices.

An operating system does the following activities

- Maintains the spooling buffer which provides a waiting station where data can rest while the slower device catches up.
- Maintains parallel computation because of spooling process as a computer can perform I/O in parallel fashion. It becomes possible to have the computer read data from a tape, write data to disk and to write out to a tape printer while it is doing its computing task.

Spooling and Buffer



Thank You
Any Questions?



پوهنتون — وڼ کاردان
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Operating System Concept Process Management Week 03

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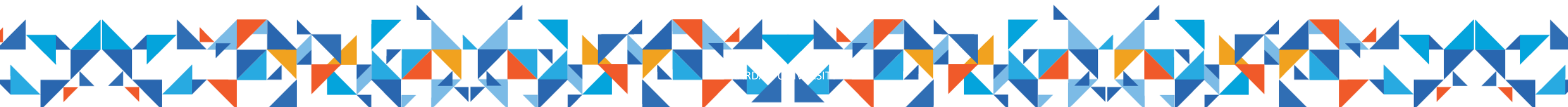


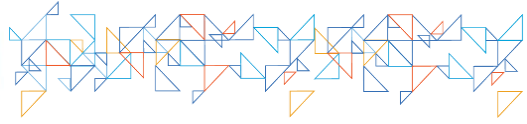
Objective

- Process Management
- Process States
- Process Control Block (PCB)
- Operations on Process
- Threads and Multithreaded Programming
- Process Synchronization



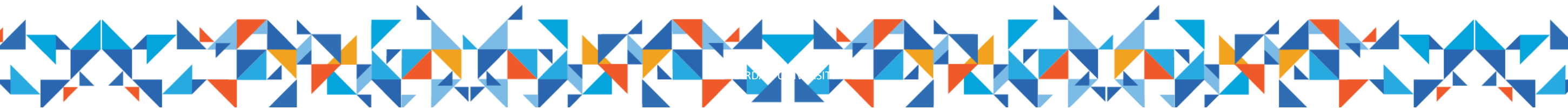
Learning Objectives





Process Management

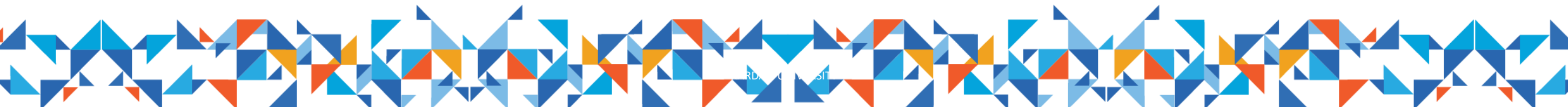
- A **process** can be thought of as a **program in execution**.
- A process will need certain resources **CPU time, memory, files, and I/O devices** to accomplish its task.
- These resources are allocated to the process either when it is **created or while it is executing**.





Cont....

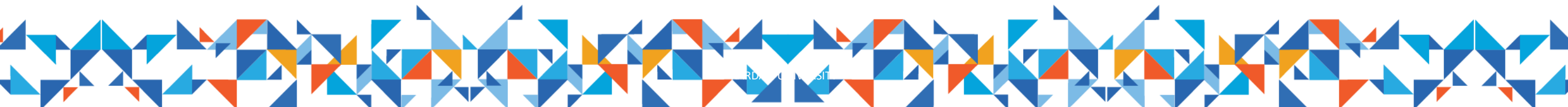
- A process is the **unit of work in most systems.**
- Systems **consist of a collection of processes:**
 1. **Operating System Process**
 2. **User Process**
- Operating-system processes **execute system code.**
- User processes **execute user code (Applications).**
- All these processes may **execute parallel.**





Cont....

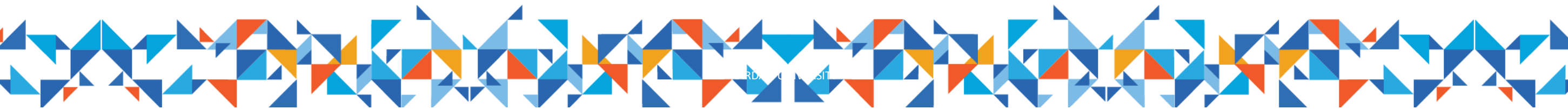
- Although traditionally a process contained only a **Single Thread** of control as it ran,
- Most modern operating systems now support processes that have multiple **Threads**.

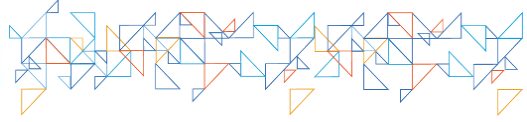




Processes

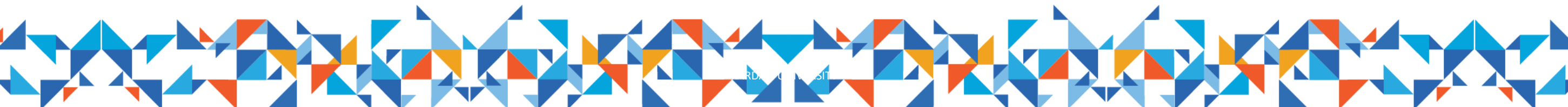
- Early computers allowed **only one program** to be **executed at a time**.
- This program **had complete control** of the system and had access to all the system's resources.
- Modern computer systems **allow multiple programs** to be **loaded into memory** and **executed concurrently**.





Process Concept

- A question that arises in discussing operating systems involves what to call all the **CPU activities?**.
- So the Answer is:
- **"Processes"** or more broadly, **"process management."**





Process State

As a process executes, it changes *state*

- ❑ **New**: The process is being created
- ❑ **Ready**: The process is waiting to be assigned to a processor
- ❑ **Running**: Instructions are being executed
- ❑ **Waiting**: The process is waiting for some event to occur
- ❑ **Terminated**: The process has finished execution

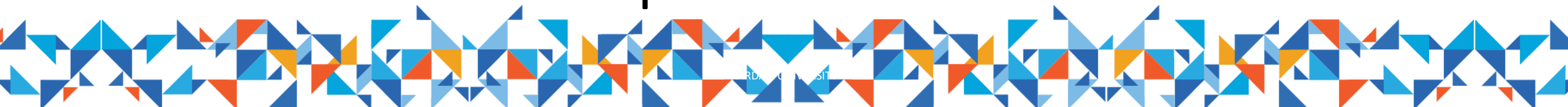
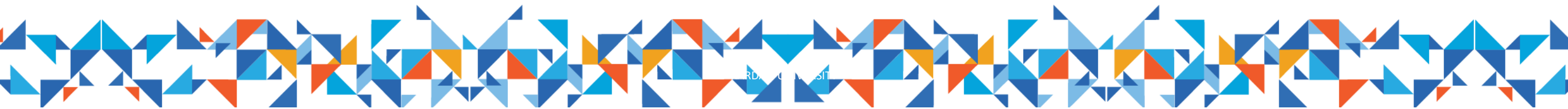
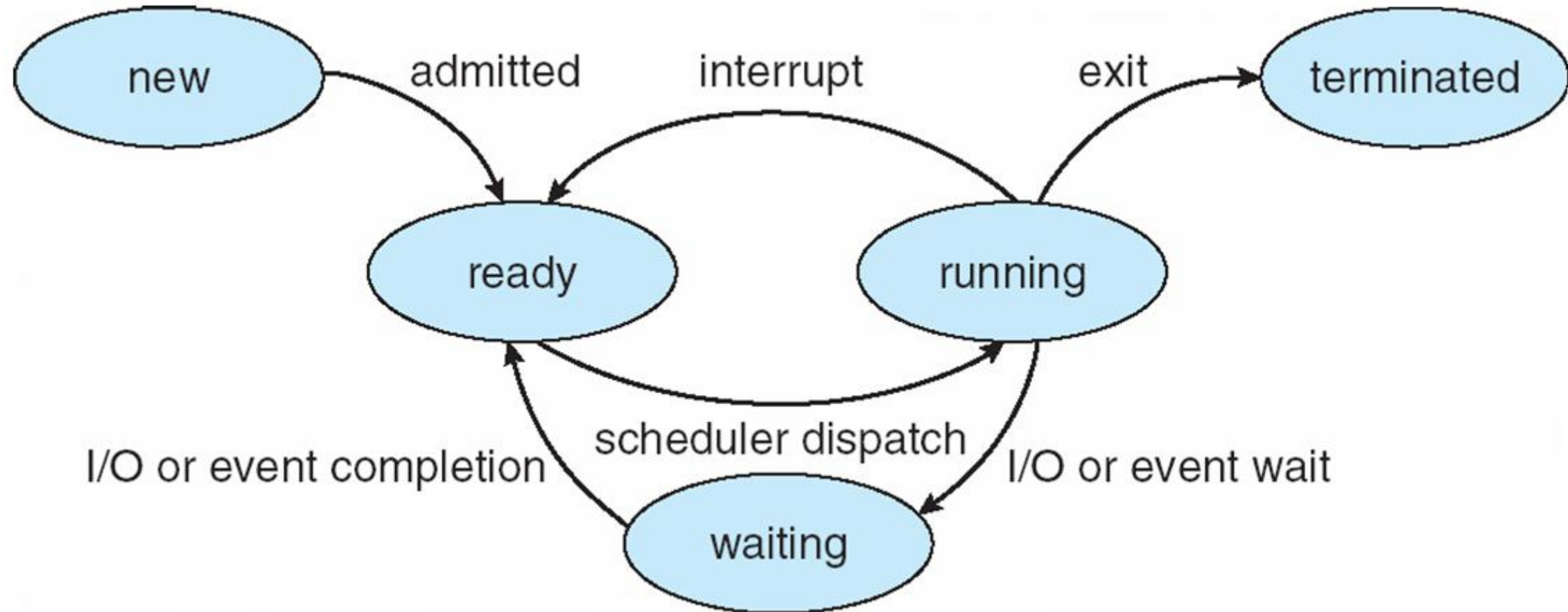




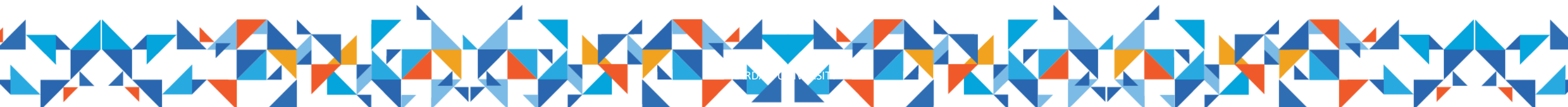
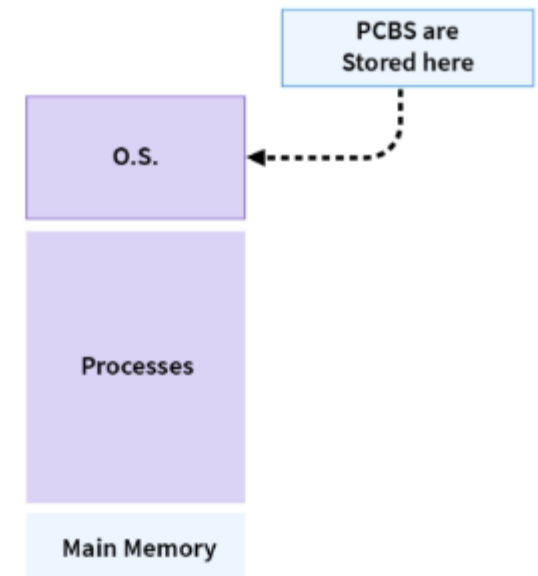
Diagram of Process State





Process Control Block (PCB)

- A **Process Control Block** is a data structure that stores information about a process such as its state, program counter, CPU registers, memory management details, and more.
- Each process has its own unique PCB.

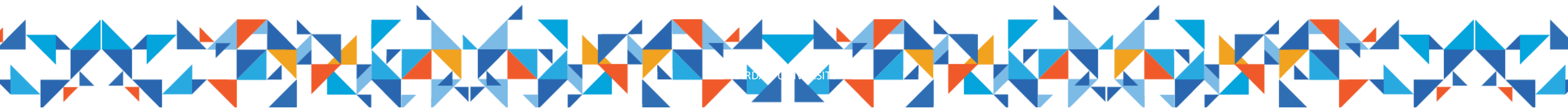


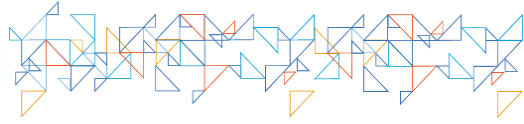


Process Control Block (PCB)

Information associated with each process

- Process state
- Program counter
- CPU registers
- CPU scheduling information
- Memory-management information
- Accounting information
- I/O status information

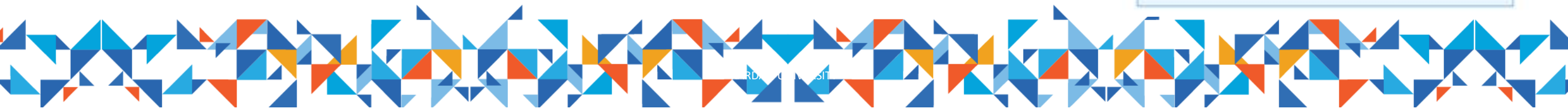




Process Control Block (PCB)

The process control block contains many attributes such as process ID, process state, process priority, accounting information, program counter, CPU registers, etc for each process.

Process ID
Process state
Process priority
Accounting information
Program counters
CPU registers
PCB pointers
List of open files
Process I/O status Information
.....





Process ID:

- When a new process is created by the user, the operating system assigns a unique ID i.e. a process-ID to that process.
- This ID helps the process to be different from other processes existing in the system.
 1. The first process will be given ID0.
 2. The second process will be given ID 1.
 3. It continues till N-1.
- These PCBs are numbered from 0 to N-1. Please note here we are giving IDs to PCBs and not to the processes.

